

Lorentz Invariance of the Klein–Gordon Field and Action

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June 6, 2026

Abstract

We study the Lorentz covariance of the Klein–Gordon equation for a real scalar field $\phi : \mathbb{M}^{1,3} \rightarrow \mathbb{R}$. Starting from the relativistic energy-momentum relation, we derive the Klein–Gordon equation and then show that its differential operator, the d’Alembertian, is invariant under Lorentz transformations. The abstract tensor argument is complemented by an explicit calculation for a boost in one spatial dimension. Finally, we verify directly that the Klein–Gordon Lagrangian density transforms as a Lorentz scalar and that the corresponding action is invariant. These results show that all inertial observers write the same field equation for the same real scalar field.

1 Introduction

The Klein–Gordon equation describes a relativistic scalar field. Mathematically, we take the real scalar field to be a smooth function

$$\phi : \mathbb{M}^{1,3} \longrightarrow \mathbb{R}, \quad (1)$$

where $\mathbb{M}^{1,3}$ denotes four-dimensional Minkowski spacetime. Thus $x \in \mathbb{M}^{1,3}$ is a spacetime event, and $\phi(x)$ is the real field value assigned to that event. In natural units, where $c = \hbar = 1$, the equation is written as

$$(\square + m^2)\phi(x) = 0. \quad (2)$$

Here

$$\square \equiv \partial_\mu \partial^\mu \quad (3)$$

is the d’Alembertian operator. We use the metric convention

$$\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1). \quad (4)$$

With this convention,

$$\square = \frac{\partial^2}{\partial t^2} - \nabla^2. \quad (5)$$

Therefore, the Klein–Gordon equation may also be written as

$$\left(\frac{\partial^2}{\partial t^2} - \nabla^2 + m^2 \right) \phi(t, \mathbf{x}) = 0. \quad (6)$$

2 Historical Note

The Klein–Gordon equation appeared in the early development of relativistic quantum mechanics as a natural attempt to combine quantum wave equations with Einstein’s special relativity. It is associated with the work of Oskar Klein and Walter Gordon, although related forms were also studied by Vladimir Fock, Erwin Schrödinger, Louis de Broglie, and others in the same period. The basic idea is to begin with the relativistic energy-momentum relation rather than the nonrelativistic expression used in the Schrödinger equation. Historically, the equation was first investigated as a candidate wave equation for a relativistic particle, but its modern interpretation is as the field equation for a spin-zero relativistic scalar field. In this field-theoretic setting, Lorentz invariance is not an optional feature: it is the mathematical statement that all inertial observers describe the same scalar field dynamics.

3 Origin from the Relativistic Energy Relation

The relativistic energy-momentum relation for a free particle of mass m is

$$E^2 = \mathbf{p}^2 + m^2. \quad (7)$$

Equivalently,

$$E^2 - \mathbf{p}^2 - m^2 = 0. \quad (8)$$

In quantum theory, energy and momentum are represented by differential operators,

$$E \longrightarrow i \frac{\partial}{\partial t}, \quad \mathbf{p} \longrightarrow -i \nabla. \quad (9)$$

Applying Eq. (9) to Eq. (8), and letting the resulting operator act on a scalar field $\phi(t, \mathbf{x})$, gives

$$\left[\left(i \frac{\partial}{\partial t} \right)^2 - (-i \nabla)^2 - m^2 \right] \phi(t, \mathbf{x}) = 0. \quad (10)$$

Since

$$\left(i \frac{\partial}{\partial t} \right)^2 = - \frac{\partial^2}{\partial t^2}, \quad (-i \nabla)^2 = - \nabla^2, \quad (11)$$

Eq. (10) becomes

$$\left(- \frac{\partial^2}{\partial t^2} + \nabla^2 - m^2 \right) \phi(t, \mathbf{x}) = 0. \quad (12)$$

Multiplying by -1 , we obtain

$$\left(\frac{\partial^2}{\partial t^2} - \nabla^2 + m^2 \right) \phi(t, \mathbf{x}) = 0, \quad (13)$$

which is precisely the Klein–Gordon equation in Eq. (6).

4 Lorentz Transformation Between Inertial Frames

Two inertial coordinate systems are related by a Lorentz transformation,

$$x'^{\mu} = \Lambda^{\mu}_{\nu} x^{\nu}. \quad (14)$$

The inverse transformation is

$$x^{\mu} = (\Lambda^{-1})^{\mu}_{\nu} x'^{\nu}. \quad (15)$$

The defining property of a Lorentz transformation is that it preserves the Minkowski metric:

$$\eta_{\alpha\beta} \Lambda^{\alpha}_{\mu} \Lambda^{\beta}_{\nu} = \eta_{\mu\nu}. \quad (16)$$

In matrix notation, this is

$$\Lambda^T \eta \Lambda = \eta. \quad (17)$$

5 Transformation of the Scalar Field

The Klein–Gordon field is a scalar field. Hence the field value at a physical spacetime event is independent of the inertial coordinate system used to describe that event:

$$\phi'(x') = \phi(x). \quad (18)$$

This equation does not say that the functional form of ϕ' and ϕ must look identical as functions of their arguments. It says that both observers assign the same field value to the same event.

6 Transformation of Derivatives

The derivative in the primed frame is found by the chain rule:

$$\begin{aligned} \partial'_{\mu} &= \frac{\partial}{\partial x'^{\mu}} \\ &= \frac{\partial x^{\nu}}{\partial x'^{\mu}} \frac{\partial}{\partial x^{\nu}}. \end{aligned} \quad (19)$$

Using Eq. (15), we obtain

$$\frac{\partial x^{\nu}}{\partial x'^{\mu}} = (\Lambda^{-1})^{\nu}_{\mu}. \quad (20)$$

Therefore,

$$\partial'_{\mu} = (\Lambda^{-1})^{\nu}_{\mu} \partial_{\nu}. \quad (21)$$

Thus spacetime derivatives transform as covectors.

7 Invariance of the d'Alembertian

In the primed inertial frame, the d'Alembertian is

$$\square' = \partial'_{\mu} \partial'^{\mu} = \eta^{\mu\nu} \partial'_{\mu} \partial'_{\nu}. \quad (22)$$

Substituting Eq. (21) into Eq. (22) gives

$$\square' = \eta^{\mu\nu} (\Lambda^{-1})^{\rho}_{\mu} (\Lambda^{-1})^{\sigma}_{\nu} \partial_{\rho} \partial_{\sigma}. \quad (23)$$

Because the inverse of a Lorentz transformation also preserves the Minkowski metric,

$$\eta^{\mu\nu}(\Lambda^{-1})^\rho{}_\mu(\Lambda^{-1})^\sigma{}_\nu = \eta^{\rho\sigma}. \quad (24)$$

Using Eq. (24) in Eq. (23), we find

$$\square' = \eta^{\rho\sigma} \partial_\rho \partial_\sigma = \square. \quad (25)$$

Thus the d'Alembertian is Lorentz invariant:

$$\boxed{\square' = \square.} \quad (26)$$

8 Explicit Check for a Boost in One Spatial Dimension

To make the previous tensor derivation more concrete, consider a boost along the x direction. The transformation is

$$t' = \gamma(t - vx), \quad (27)$$

$$x' = \gamma(x - vt), \quad (28)$$

where

$$\gamma = \frac{1}{\sqrt{1 - v^2}}. \quad (29)$$

The inverse transformation is

$$t = \gamma(t' + vx'), \quad (30)$$

$$x = \gamma(x' + vt'). \quad (31)$$

By the chain rule,

$$\begin{aligned} \frac{\partial}{\partial t'} &= \frac{\partial t}{\partial t'} \frac{\partial}{\partial t} + \frac{\partial x}{\partial t'} \frac{\partial}{\partial x} \\ &= \gamma \frac{\partial}{\partial t} + \gamma v \frac{\partial}{\partial x}, \end{aligned} \quad (32)$$

$$\begin{aligned} \frac{\partial}{\partial x'} &= \frac{\partial t}{\partial x'} \frac{\partial}{\partial t} + \frac{\partial x}{\partial x'} \frac{\partial}{\partial x} \\ &= \gamma v \frac{\partial}{\partial t} + \gamma \frac{\partial}{\partial x}. \end{aligned} \quad (33)$$

Therefore,

$$\frac{\partial^2}{\partial t'^2} = \gamma^2 \left(\frac{\partial^2}{\partial t^2} + 2v \frac{\partial^2}{\partial t \partial x} + v^2 \frac{\partial^2}{\partial x^2} \right), \quad (34)$$

$$\frac{\partial^2}{\partial x'^2} = \gamma^2 \left(v^2 \frac{\partial^2}{\partial t^2} + 2v \frac{\partial^2}{\partial t \partial x} + \frac{\partial^2}{\partial x^2} \right). \quad (35)$$

Subtracting Eq. (35) from Eq. (34), the mixed derivative terms cancel:

$$\begin{aligned} \frac{\partial^2}{\partial t'^2} - \frac{\partial^2}{\partial x'^2} &= \gamma^2(1 - v^2) \left(\frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2} \right) \\ &= \frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2}. \end{aligned} \quad (36)$$

Thus, in 1 + 1 dimensions,

$$\boxed{\frac{\partial^2}{\partial t'^2} - \frac{\partial^2}{\partial x'^2} = \frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2}}. \quad (37)$$

This is the explicit version of the invariance of the d'Alembertian.

9 Invariance of the Klein–Gordon Equation

In the primed frame, the Klein–Gordon equation has the form

$$(\square' + m^2)\phi'(x') = 0. \quad (38)$$

Using the scalar-field transformation in Eq. (18) and the d'Alembertian invariance in Eq. (25), we have

$$(\square' + m^2)\phi'(x') = (\square + m^2)\phi(x). \quad (39)$$

If the field satisfies the Klein–Gordon equation in the unprimed frame,

$$(\square + m^2)\phi(x) = 0, \quad (40)$$

then Eq. (39) implies

$$(\square' + m^2)\phi'(x') = 0. \quad (41)$$

Therefore the Klein–Gordon equation has the same form in both inertial frames.

10 Invariant Lagrangian Formulation

The free real Klein–Gordon field can also be described by the Lagrangian density

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2. \quad (42)$$

To check the invariance of the Lagrangian density, write the transformed density in the primed frame:

$$\mathcal{L}'(x') = \frac{1}{2}\partial'_\mu\phi'(x')\partial'^\mu\phi'(x') - \frac{1}{2}m^2\phi'(x')^2. \quad (43)$$

Since ϕ is a scalar field, Eq. (18) implies

$$\phi'(x')^2 = \phi(x)^2. \quad (44)$$

For the derivative of the field, we use the chain rule together with the scalar transformation:

$$\begin{aligned} \partial'_\mu\phi'(x') &= \frac{\partial x^\nu}{\partial x'^\mu}\partial_\nu\phi(x) \\ &= (\Lambda^{-1})^\nu{}_\mu\partial_\nu\phi(x). \end{aligned} \quad (45)$$

The kinetic term in the primed frame is therefore

$$\begin{aligned} \partial'_\mu\phi'(x')\partial'^\mu\phi'(x') &= \eta^{\mu\lambda}\partial'_\mu\phi'(x')\partial'_\lambda\phi'(x') \\ &= \eta^{\mu\lambda}(\Lambda^{-1})^\nu{}_\mu(\Lambda^{-1})^\rho{}_\lambda\partial_\nu\phi(x)\partial_\rho\phi(x). \end{aligned} \quad (46)$$

Using the metric-preservation identity for the inverse Lorentz transformation,

$$\eta^{\mu\lambda}(\Lambda^{-1})^\nu{}_\mu(\Lambda^{-1})^\rho{}_\lambda = \eta^{\nu\rho}, \quad (47)$$

Eq. (46) becomes

$$\partial'_\mu \phi'(x') \partial'^\mu \phi'(x') = \eta^{\nu\rho} \partial_\nu \phi(x) \partial_\rho \phi(x) = \partial_\nu \phi(x) \partial^\nu \phi(x). \quad (48)$$

Combining Eq. (44) and Eq. (48), we obtain

$$\begin{aligned} \mathcal{L}'(x') &= \frac{1}{2} \partial'_\mu \phi'(x') \partial'^\mu \phi'(x') - \frac{1}{2} m^2 \phi'(x')^2 \\ &= \frac{1}{2} \partial_\mu \phi(x) \partial^\mu \phi(x) - \frac{1}{2} m^2 \phi(x)^2 \\ &= \mathcal{L}(x). \end{aligned} \quad (49)$$

Thus the Lagrangian density is a Lorentz scalar:

$$\boxed{\mathcal{L}'(x') = \mathcal{L}(x)}. \quad (50)$$

The action is

$$S = \int d^4x \mathcal{L}. \quad (51)$$

For a proper Lorentz transformation, the four-volume element is invariant,

$$d^4x' = |\det \Lambda| d^4x = d^4x, \quad (52)$$

because $|\det \Lambda| = 1$. Therefore,

$$S' = \int d^4x' \mathcal{L}'(x') = \int d^4x \mathcal{L}(x) = S. \quad (53)$$

The action formulation therefore gives another way to see the Lorentz invariance of the Klein–Gordon theory.

11 Conclusion

The Klein–Gordon field equation is invariant under Lorentz transformations because the scalar field satisfies Eq. (18) and the d’Alembertian satisfies Eq. (26). The same structure is visible in the explicit boost calculation Eq. (37) and in the invariant action Eq. (53). Hence

$$(\partial_\mu \partial^\mu + m^2)\phi = 0 \quad (54)$$

is a Lorentz-covariant equation: every inertial observer writes the same equation for the same physical scalar field.